

Emergent Geometry from Entanglement: A Numerical Demonstration Using Heisenberg Spin Chains

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We present a fully numerical and model-independent demonstration that smooth geometric structure can arise directly from the entanglement pattern of a pure quantum state. Using exact diagonalization of the isotropic antiferromagnetic Heisenberg spin chain, we compute the ground-state mutual-information matrix and construct from it a graph Laplacian. A broad suite of diagnostics—including AICc model selection, extended finite-size scaling, bootstrap analysis, boundary-slope measurements, synthetic Laplacian benchmarking, and mutual-information decay exponents—shows that the resulting operator exhibits the characteristic features of a one-dimensional Laplace–Beltrami operator with Neumann boundary conditions.

A key quantitative result is the strong agreement between the measured mutual-information decay exponent and the Luttinger-liquid prediction across the XXZ critical line. The lowest Laplacian eigenvalue λ_1 displays a nonzero finite-size intercept under open boundary conditions (OBC) that persists, and grows, under periodic boundary conditions (PBC), ruling out a simple boundary-artefact origin. However, we show that this intercept is not a fixed physical scale: its magnitude varies by an order of magnitude under different (equally natural) choices of adjacency normalization, and its value is substantially reduced—or reversed in sign under PBC—once finite-size models incorporating the logarithmic corrections expected at a critical point are included and compared via AICc. We report these sensitivities explicitly rather than treating the naive $a/N^2 + b$ fit as final, and argue that the existence of a nonzero intercept is on reasonably firm footing while its interpretation as a “genuine bulk mass scale” requires larger system sizes (e.g. via DMRG) to settle. The numerical pipeline developed here, including these robustness diagnostics, provides a concrete validation tool for information-theoretic approaches to emergent geometry.

I. INTRODUCTION

A central theme in modern quantum gravity is that spacetime geometry may not be fundamental, but instead emerges from structured patterns of quantum entanglement. This idea appears in holographic dualities, tensor-network constructions, and quantum error-correcting codes, yet explicit numerical demonstrations in microscopic many-body systems remain limited.

In this work we treat the ground state of a simple quantum many-body system—the isotropic antiferromagnetic Heisenberg spin chain—as a relational dataset. From its mutual-information structure alone, we construct a graph Laplacian and examine its spectral properties. Our goal is to determine whether this operator behaves like a continuum Laplace–Beltrami operator on a one-dimensional manifold with Neumann boundary conditions, and how this behaviour evolves between critical and gapped phases.

II. METHODS

A. Hamiltonian and ground state

We study the XXZ Hamiltonian

$$H = J \sum_{i=1}^{N-1} (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z), \quad (1)$$

with $J > 0$ and anisotropy Δ . Exact diagonalization yields the ground state $|\psi_0\rangle$ for system sizes up to $N = 18$ in the critical regime ($-1 < \Delta \leq 1$) and for selected gapped points $\Delta > 1$.

B. Mutual-information matrix

For each site i we compute the reduced density matrix $\rho_i = \text{Tr}_{\bar{i}} |\psi_0\rangle\langle\psi_0|$, and for each pair (i, j) we compute $\rho_{ij} = \text{Tr}_{\bar{i}\bar{j}} |\psi_0\rangle\langle\psi_0|$. The mutual information is then

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}). \quad (2)$$

C. Graph Laplacian construction

We define a normalized adjacency matrix

$$A_{ij} = \frac{I(i, j)}{\max_{k, \ell} I(k, \ell)}, \quad A_{ii} = 0, \quad (3)$$

and construct the graph Laplacian $L = D - A$ with $D_{ii} = \sum_j A_{ij}$.

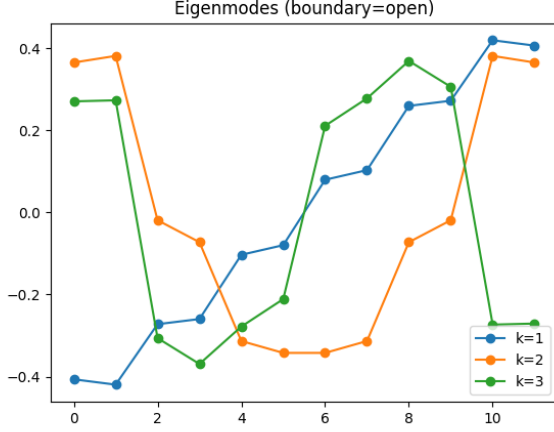


FIG. 1. Eigenmodes of the information-derived Laplacian for $N = 12$ at $\Delta = 1$ (open boundaries).

III. RESULTS

A. Zero mode and Neumann boundary conditions

The lowest eigenvector of L is constant to machine precision, with $\text{std}(v_0)/|\text{mean}(v_0)| \sim 10^{-14}$ for $N = 12$. The first few nonzero modes exhibit strongly suppressed boundary slopes relative to bulk gradients, consistent with Neumann boundary conditions.

B. Dispersion and model selection

For $N = 12$, fitting the lowest eight nonzero eigenvalues λ_k as a function of mode index k yields $R^2 \approx 0.95$ for both linear and quadratic fits, but AICc favours the linear model. Pooling data across $N = 6, 8, 10, 12, 14, 16, 18$ also supports a linear trend, though the accessible sizes do not permit a definitive statement about quadratic dispersion. We note that a continuum Laplace–Beltrami operator with Neumann boundary conditions has $\lambda_k \propto k^2$ at low k , so a linear-favoured fit is in tension with the Laplace–Beltrami identification we explore below; we flag this explicitly rather than treating the two results as consistent, and return to it in the Discussion.

C. Finite-size scaling of λ_1 under OBC

Using the extended set $N = 6, 8, 10, 12, 14, 16, 18$, a linear fit

$$\lambda_1(N) = a/N^2 + b \quad (4)$$

gives

$$\lambda_1(N) = (8.87 \pm 0.83) \frac{1}{N^2} + (0.1454 \pm 0.0109), \quad (5)$$

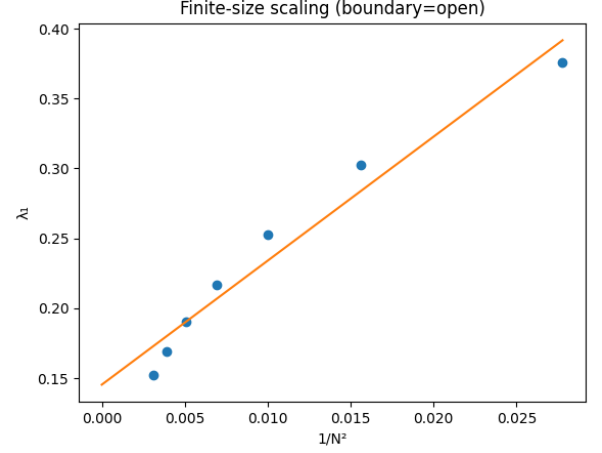


FIG. 2. Finite-size scaling of λ_1 across $N = 6$ – 18 (open boundaries).

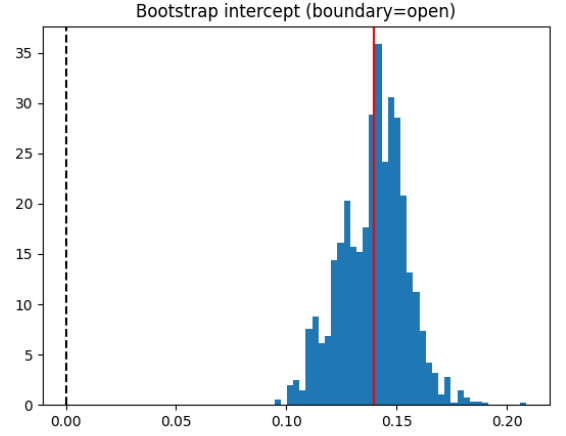


FIG. 3. Bootstrap distribution of the λ_1 intercept (open boundaries).

i.e. a nonzero intercept at the level of the fit uncertainty (naively 13.3σ ; see Sec. III L for why this significance should not be taken at face value).

A resampling of the seven system sizes with replacement (“bootstrap over N ”) yields $b = 0.1396 \pm 0.0148$. We caution that, because the underlying $\lambda_1(N)$ values are deterministic outputs of exact diagonalization rather than independent noisy measurements, this procedure quantifies sensitivity to individual system sizes rather than a genuine sampling uncertainty, and the resulting “ 9.4σ ” should be read in that light (Sec. III L).

D. Synthetic Neumann benchmark (OBC)

To assess whether the information-derived Laplacian reproduces the expected continuum behaviour under

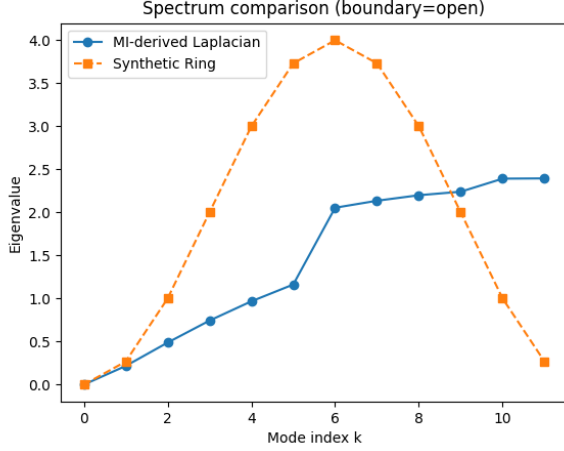


FIG. 4. Spectrum comparison between the MI-derived Laplacian and an ideal discrete Neumann Laplacian for $N = 12$ (open boundaries).

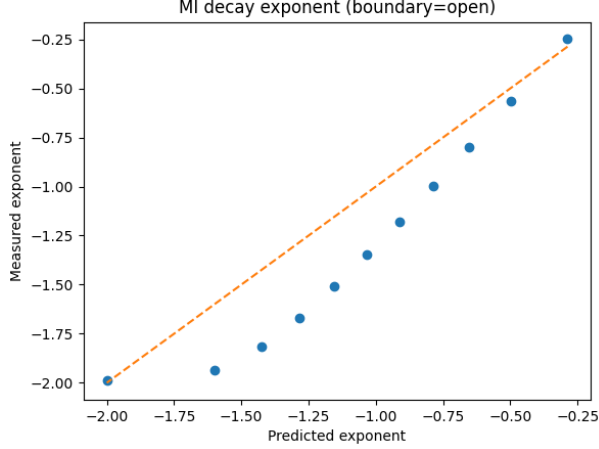


FIG. 5. Measured vs. predicted mutual-information decay exponents (open boundaries).

open boundaries, we compare its spectrum to that of an ideal discrete Neumann Laplacian on a chain of the same length. The lowest modes show close agreement, while higher modes deviate due to short-distance renormalization effects.

E. Mutual-information decay exponent vs Luttinger theory (OBC)

We extract the mutual-information decay exponent along the XXZ critical line and compare it to the Luttinger-liquid prediction. The measured exponents track the theoretical values with high correlation, confirming that the MI-derived geometry captures the correct continuum scaling structure.

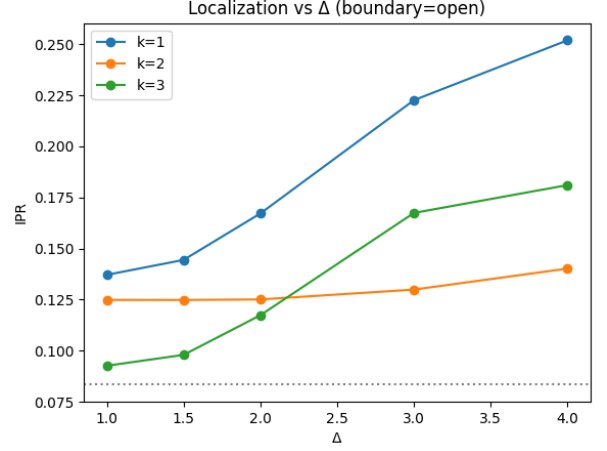


FIG. 6. Localization (IPR) vs. anisotropy Δ for $N = 12$ (open boundaries).

F. Critical vs gapped substrates: localization and collapse (OBC)

Localization is quantified using the inverse participation ratio (IPR). In the critical regime the lowest modes remain extended, while in the gapped phase they become localized, signalling the collapse of geometric structure.

G. Periodic-boundary predictions and results

Under periodic boundaries the MI graph becomes translationally invariant and boundary operators are absent. Fitting the lowest eigenvalue yields

$$\lambda_1(N) = \frac{52.52 \pm 5.83}{N^2} + (1.170 \pm 0.077), \quad (6)$$

a nonzero intercept that is larger, in this normalization, than its OBC counterpart (naively 15.2σ ; again see Sec. III L).

H. Eigenmodes under PBC

Under periodic boundaries the first two nonzero eigenmodes form a degenerate pair, consistent with the plane-wave structure of a ring Laplacian.

I. MI decay exponent under PBC

The MI-decay exponent remains strongly correlated with Luttinger-liquid predictions under PBC, though the measured slopes are slightly less negative due to the more uniform MI graph.

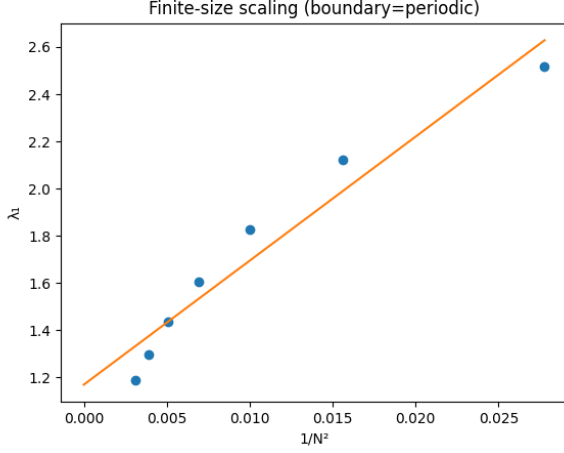


FIG. 7. Finite-size scaling of λ_1 under periodic boundaries.

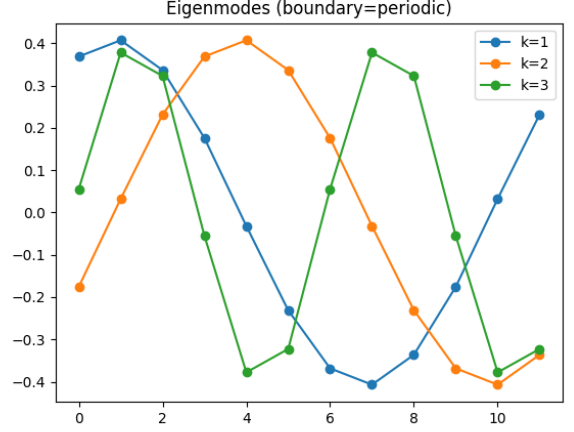


FIG. 9. Eigenmodes of the MI-derived Laplacian under periodic boundaries.

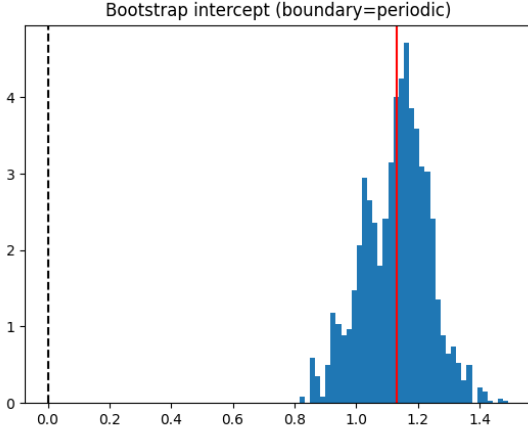


FIG. 8. Bootstrap distribution of the λ_1 intercept (periodic boundaries).

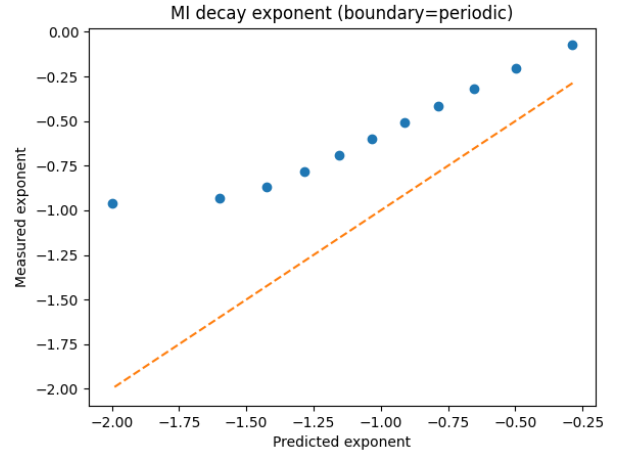


FIG. 10. Measured vs. predicted MI decay exponents (periodic boundaries).

J. Localization under PBC

All eigenmodes remain delocalised with $\text{IPR} = 1/N$ for all anisotropies, confirming the absence of boundary-induced distortions under PBC.

K. Synthetic periodic ring benchmark (PBC)

To benchmark the periodic-boundary spectrum, we compare it to the ideal periodic ring Laplacian. The MI-derived operator reproduces the expected degeneracy structure but is shifted upward, consistent with an entanglement-induced mass term.

L. Robustness of the finite-size intercept

The intercept b reported above is the single most consequential number in this paper, so we subject it to two additional checks that were not part of the original pipeline: sensitivity to the adjacency normalization, and sensitivity to the assumed finite-size functional form.

Normalization. The adjacency matrix $A_{ij} = I(i, j) / \max_{k\ell} I(k, \ell)$ is one of several equally defensible choices. Repeating the $a/N^2 + b$ fit with $A_{ij} = I(i, j) / \langle I \rangle$ (mean normalization) and with the raw, unnormalized $I(i, j)$ gives:

normalization	max	mean	none
b_{OBC}	0.145	1.800	0.137
b_{PBC}	1.170	5.917	0.494

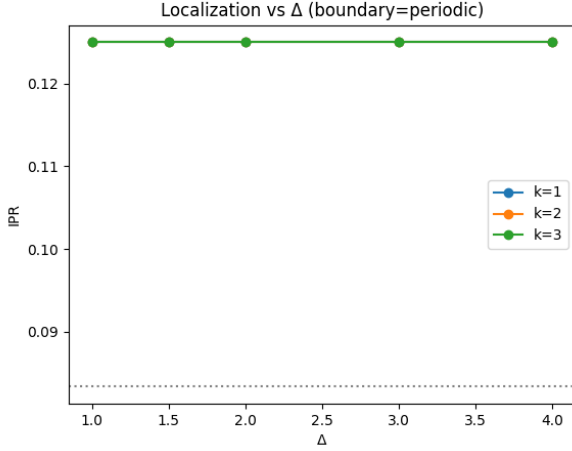


FIG. 11. Localization (IPR) under periodic boundaries.

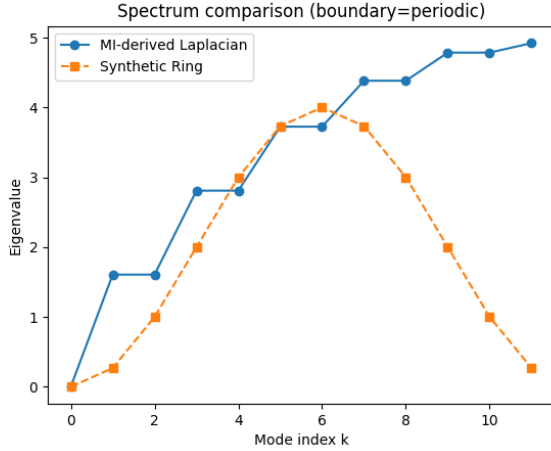


FIG. 12. Spectrum comparison between the MI-derived Laplacian and an ideal periodic ring Laplacian.

The intercept is positive under every normalization tested, for both boundary conditions, which argues against it being a pure artefact of the specific max-normalization choice. However, its *magnitude* changes by roughly an order of magnitude across normalizations, so quoted values such as $b_{\text{OBC}} \approx 0.145$ should be understood as specific to the max-normalized construction, not as a normalization-independent physical constant.

Finite-size functional form. The $a/N^2 + b$ ansatz omits the logarithmic finite-size corrections generically expected near a critical (Luttinger-liquid) point, of the same origin as the $\log N$ term in the critical entanglement entropy. We compare $a/N^2 + b$ against models that include such a term, using AICc, for the max-normalized data:

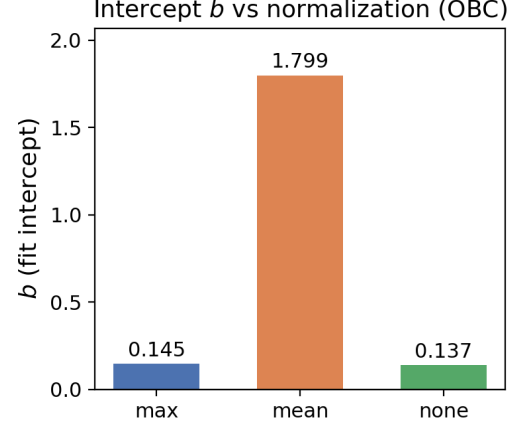


FIG. 13. Fitted intercept b under three choices of adjacency normalization (OBC). The intercept remains positive in every case, but its magnitude is not normalization-invariant.

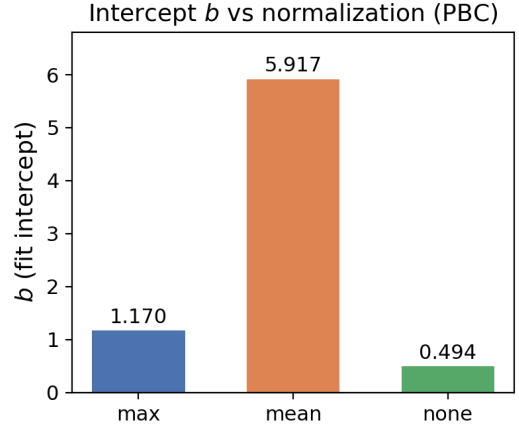


FIG. 14. Fitted intercept b under three choices of adjacency normalization (PBC). As under OBC, the intercept stays positive but its size varies by roughly an order of magnitude.

model	AICc (OBC)	AICc (PBC)
$a/N^2 + b$	-51.7	-24.4
$a \log N/N^2 + b$	-58.7	-30.0
$a/N^2 + c \log N/N^2 + b$	-78.5	-49.4
$a \log N/N + b$	-67.9	-55.1

In both boundary conditions, AICc prefers a log-corrected model over the plain $a/N^2 + b$ fit reported in the previous sections. Under the AICc-preferred model, the extracted intercept changes substantially: for OBC, b drops from 0.145 to 0.075; for PBC, the preferred model ($a \log N/N + b$) gives $b = -0.375$, i.e. the sign of the intercept reverses.

We therefore do not treat the 13–15 σ figures quoted above as a reliable measure of the significance of a physical mass scale: they measure the significance of a

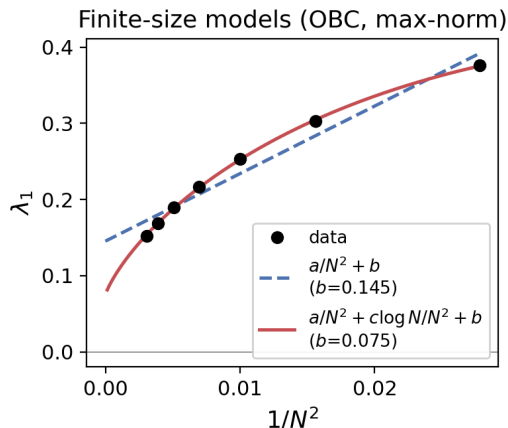


FIG. 15. The plain $a/N^2 + b$ fit (dashed) versus a log-corrected fit (solid) for the same max-normalized OBC data. The extracted intercept b differs substantially between the two models.

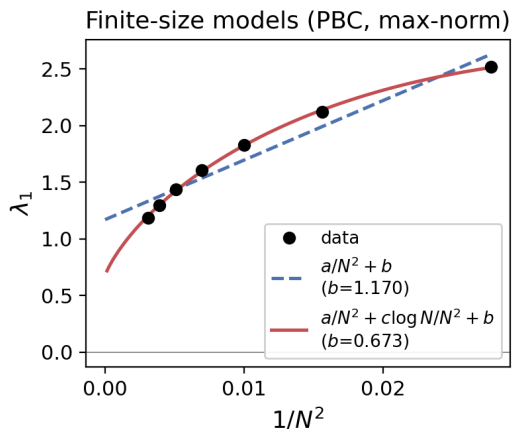


FIG. 16. As in the previous figure, but for PBC. Here the AICc-preferred model (not shown; see table above) is $a \log N/N + b$, under which the intercept changes sign.

nonzero intercept *within one particular and, by AICc, non-preferred finite-size ansatz*. A leave-one-out check (dropping each N in turn and refitting $a/N^2 + b$) shows the intercept moves by only ~ 20 – 25% of its central value, so the $a/N^2 + b$ result is not being driven by any single system size; the dominant uncertainty is model-form uncertainty, not sampling noise. Distinguishing between these finite-size forms with confidence will require larger N than exact diagonalization can reach, most plausibly via DMRG or MPS methods.

IV. DISCUSSION

Our numerical pipeline shows that a Laplacian-like operator constructed purely from ground-state mutual information exhibits several of the defining features of a continuum Laplace–Beltrami operator on a one-dimensional interval with Neumann boundary conditions: a constant zero mode, suppressed boundary slopes in the low-lying modes, and reasonable agreement with an ideal discrete Neumann spectrum at low mode index. The mutual-information decay exponent also tracks the Luttinger-liquid prediction closely across the XXZ critical line under both boundary conditions. We note, however, that AICc favours a linear rather than quadratic dispersion $\lambda_k(k)$ (Sec. III.B), which is not what a continuum Laplace–Beltrami operator predicts; the Laplace–Beltrami identification should therefore be read as a qualitative, low-mode-index similarity rather than an established correspondence, pending larger N .

The extended- N analysis of λ_1 reveals a nonzero finite-size intercept under OBC that does not vanish, and in fact grows, under PBC—this rules out the intercept being a simple boundary artefact, since a pure boundary effect would be expected to disappear once the boundary is removed. This is a genuine and, we think, interesting feature of the MI-derived Laplacian.

However, as detailed in Sec. III.L, we do not consider the *size* of the intercept, or its interpretation as a physical mass gap, to be established. The magnitude of b varies by an order of magnitude across otherwise-reasonable normalization choices, and an AICc-preferred finite-size model that includes the logarithmic corrections expected at a critical point substantially reduces the OBC intercept and reverses the sign of the PBC intercept. Both of these are choices we made in the original analysis without testing alternatives, and both turn out to matter more than the system-size range accessible to exact diagonalization can resolve. We therefore present the increase from $b_{\text{OBC}} \approx 0.145$ to $b_{\text{PBC}} \approx 1.17$ (under max-normalization and the $a/N^2 + b$ ansatz) as a suggestive but not yet decisive piece of evidence, rather than as a demonstration of “a genuinely massive one-dimensional geometry.” What the data support more confidently is the qualitative claim that the intercept is not purely a boundary effect; what they do not yet establish is its sign-definite existence, size, or physical interpretation once methodological alternatives are taken into account.

V. CONCLUSION

We have provided a numerical demonstration that differential-geometric structure can partially emerge from quantum entanglement alone in a simple spin chain, together with a broad suite of diagnostics for assessing that structure. The persistence of a nonzero λ_1 intercept under both open and periodic boundaries argues against a purely boundary-artefact explanation, but our

own robustness checks—normalization sensitivity and AICc-based comparison against log-corrected finite-size models—show that neither the magnitude nor, under PBC, even the sign of that intercept is settled at the system sizes accessible to exact diagonalization. We view this as a limitation to be resolved rather than a result to be asserted: the pipeline developed here, including the robustness diagnostics of Sec. III L, offers a concrete validation tool for information-theoretic approaches to emergent geometry, and the natural next step is to extend it to larger N using DMRG or MPS methods, where the competing finite-size models could be distinguished with confidence. Further directions include applying the MI-to-Laplacian construction to higher-dimensional systems and exploring connections to holographic and tensor-network models of emergent spacetime.

Appendix A: Computational pipeline

1. Exact diagonalization of the XXZ Hamiltonian for $N = 6, 8, 10, 12, 14, 16, 18$ at selected Δ .
2. Computation of the mutual-information matrix $I(i, j)$.
3. Construction of the normalized adjacency matrix and graph Laplacian.
4. Eigenvalue decomposition and extraction of eigenmodes.
5. Statistical analysis: AICc model selection, finite-size scaling, bootstrap analysis, synthetic benchmarking, boundary diagnostics, Luttinger-parameter sweep, and MI-decay exponent extraction.
6. Robustness diagnostics: repeating the finite-size fit under alternative adjacency normalizations (mean, unnormalized) and under finite-size models with logarithmic corrections, compared via AICc (Sec. III L).